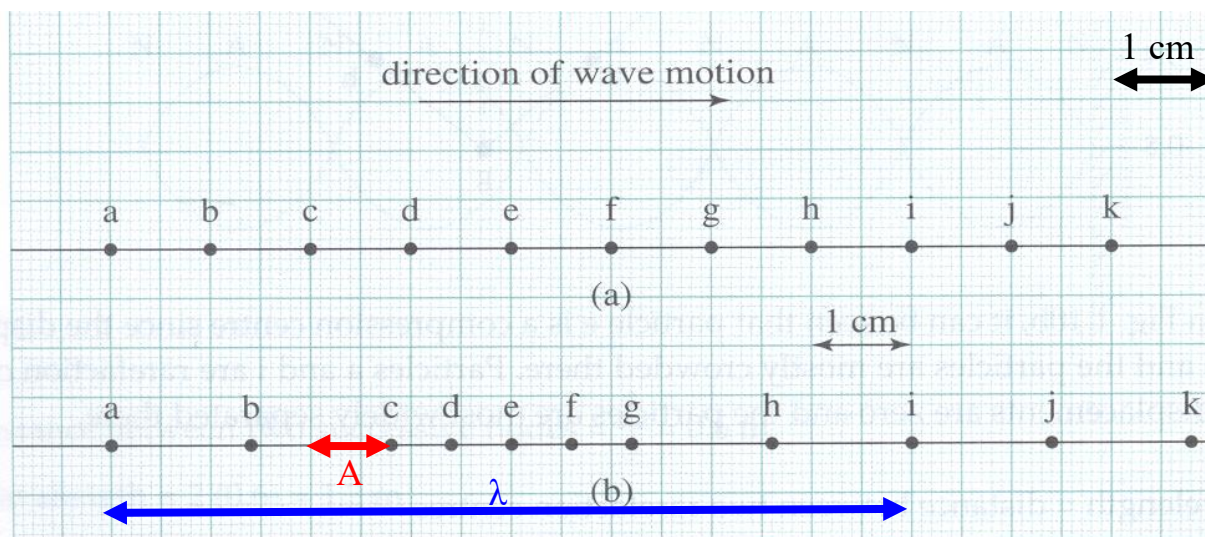
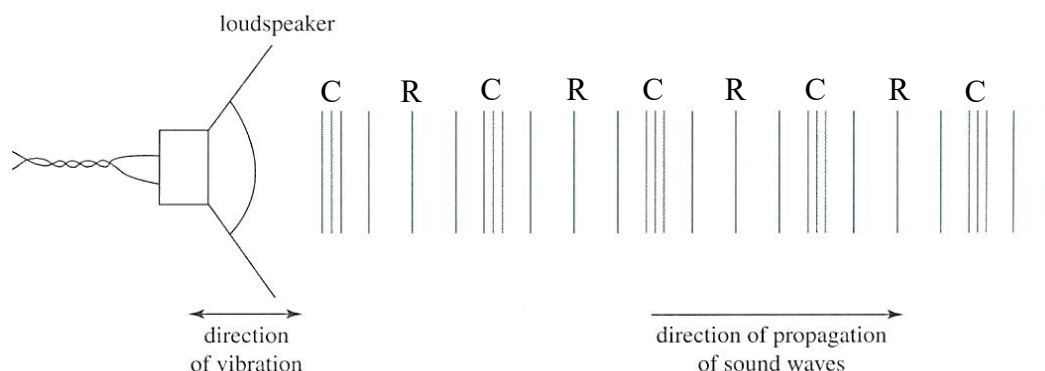


Sound waves

Sound waves are mechanical **longitudinal waves** which consist of compressions and rarefactions. Direction of oscillation of a sound particle in the medium is **parallel** to the direction of propagation



Rarefaction

Compression

Rarefaction

Amplitude (A) is the maximum displacement of any particle from its equilibrium position.

Wavelength (λ) is the distance between two successive compressions or rarefactions.

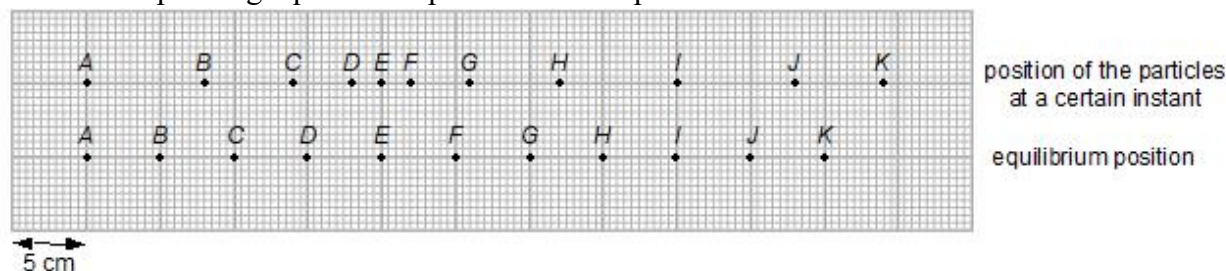
1. What is the amplitude of particle c?

Answer : 0.8 cm

2. What is the wavelength of the wave?

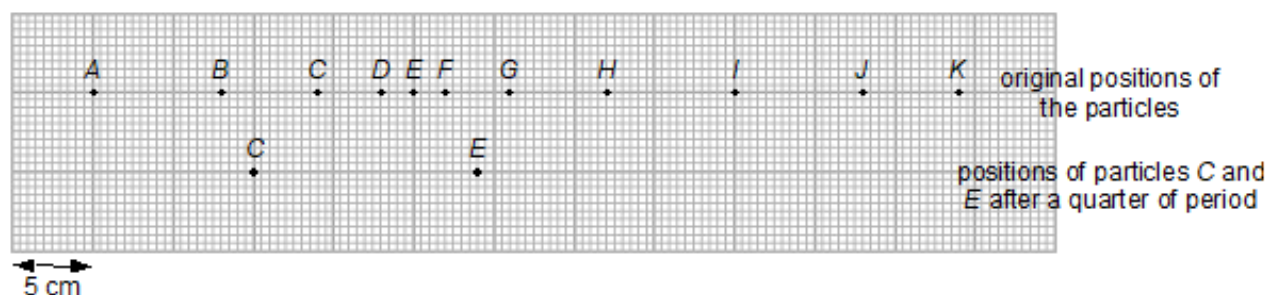
Answer : 8 cm

3. The figures below show the positions of some particles in a medium when a wave is passing through them, and the corresponding equilibrium positions of the particles.



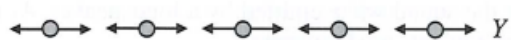
- State the type of wave travelling in the medium.
- Which particle(s) is/are
 - at the centre of a compression?
 - at the centre of a rarefaction?
 - at rest momentarily?
- What is the wavelength of the wave?
- It is known that each of the particles takes 0.2 s to complete one oscillation. Find the
 - frequency,
 - speed,
 - amplitude of the wave.
- Suppose the wave is travelling to the right. Draw the positions of particles C and E after a quarter of a period.

- Longitudinal wave (1A)✓
- Particle E (1A)✓
 - Particles A and I (1A)✓
 - Particles C and K (1A)✓
- The distance between particles E and I is equal to half a wavelength (1M).
Therefore, the wavelength is $\lambda = 2 \times 20 = 40$ cm (1A)✓
- Frequency of the wave = $1/0.2 = 5$ Hz (1M+1A)✓
 - Speed of the wave = $f\lambda = 5 \times 40 = 200$ m s⁻¹ (1M+1A)✓
 - The amplitude of the wave is 4 cm. (1A)✓
- ✓



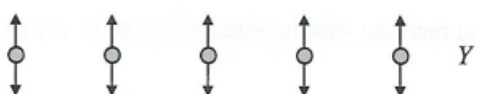
(1A)✓

4. A sound wave is produced at X and reaches Y. Which of the following diagrams correctly shows the motion of the air particles between X and Y? (The direction of the arrows shows the direction of motion of the air particles)

A X  Y

B X  Y

C X  Y

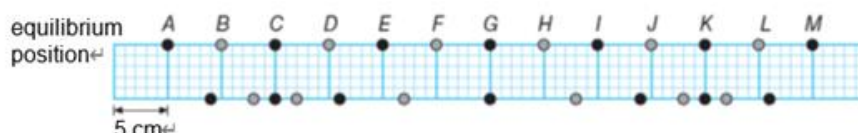
D X  Y

Sound wave is a longitudinal wave. The air particles vibrate about their equilibrium positions along the direction of propagation.

A

5. The figure below shows a longitudinal wave travelling through some particles. What is the amplitude of the wave?

- A 2 cm
B 3 cm
C 4 cm
D 5 cm

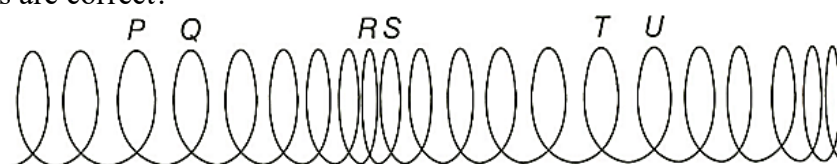


C

6. A longitudinal wave travels along a slinky spring. The figure below shows the waveform at a certain instant. Which of the following statements are correct?

- (1) P and R vibrate in antiphase.
(2) Q and U vibrate in phase.
(3) T is momentarily at rest.

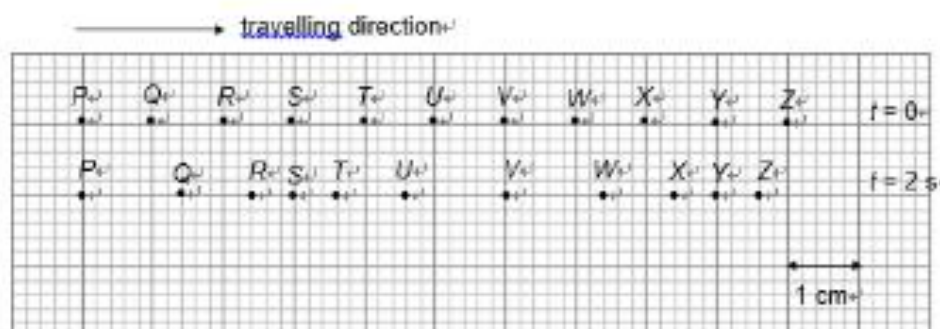
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. All of the above



A

7. The following figure shows a longitudinal wave. Which of the following pair of particles are in antiphase?

- A P and V
B S and Y
C R and T
D V and Y



D

Demonstration the longitudinal nature of sound wave



- Place a candle with flame in front of a loudspeaker
- When the loudspeaker transmits sound, the candle flame is found to oscillate to and fro

Comparisons between sound and light

	Sound waves	Light waves
Wave nature	● Both of them demonstrate reflection, refraction, diffraction and interference. ● Both of them obey the wave equation '$v = f\lambda$' ● Sound is a mechanical wave but light is an electromagnetic wave.	
Direction of vibration	Longitudinal wave Vibration is parallel to the travelling direction of the wave	Transverse wave Vibration is perpendicular to the travelling direction of the wave
Medium	● Requires a medium for transmission. ● Cannot travel through vacuum.	Can travel through vacuum or any transparent medium.
Speed in air	● $330 - 340 \text{ ms}^{-1}$ in air ● $V_{\text{solid}} > V_{\text{liquid}} > V_{\text{gas}}$	● $3 \times 10^8 \text{ ms}^{-1}$ in a vacuum or air ● $V_{\text{vacuum}} > V_{\text{gas}} > V_{\text{liquid}} > V_{\text{solid}}$
Frequency	● Audible sound: 20-20 kHz ● Ultrasound: above 20 kHz	Visible light: $\sim 10^{14} \text{ Hz}$
Wavelength	● Audible sound: $10^{-2} \text{ m} - 10^1 \text{ m}$ ● Diffraction of sound waves is very obvious.	● Visible light: 10^{-7} m ● Diffraction of light wave is hard to observe.

8. Which of the following statements about sound waves is correct?

- A They are transverse waves.
- B They are electromagnetic waves.
- C They travel at a speed of $3 \times 10^8 \text{ m s}^{-1}$ in air.
- D They cannot travel through a vacuum.

D

9. Which statements about ultrasound are correct ?

- (1) Ultrasound is longitudinal waves.
- (2) Ultrasound needs media to propagate.
- (3) The speed of ultrasound in glass is higher than that in air.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

D

10. Which of the following are the key evidences showing that sound is a kind of wave?
- (1) When sound is reflected, its angle of incidence equals its angle of reflection.
 - (2) We can hear people talking around a corner.
 - (3) Alternative loud and soft sounds are heard when we walk across two separate loudspeakers playing the same note.

- A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

C

11. Which of the following statements about light and sound is correct?

- A Both light and sound are transverse waves.
B Both light and sound can travel in a vacuum.
C Both light and sound travel faster in air than in water.
D Both light and sound undergo refraction when travelling from one medium to another

D

12. Which of the following statements about light waves and sound waves is/are correct ?

- (1) Light waves are transverse while sound waves are longitudinal
- (2) Light waves are propagating but sound waves are not
- (3) Both waves travel at the same speed in vacuum

- A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

- (1) ✓ Light wave is transverse while sound wave is longitudinal.
(2) × Both waves are travelling, i.e. energy is transmitted when the waves propagate.
(3) × Sound wave requires a medium for transmission and so it cannot travel in vacuum.

A

13. Which of the following statements comparing a sound wave and an electromagnetic wave must be correct?

- (1) Their speeds decrease when they travel from air to glass.
- (2) The wavelength of a sound wave in air is larger than that of an electromagnetic wave.
- (3) Both waves transfer energy.

- A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. All of the above

Sound travels faster in solid than in gas.

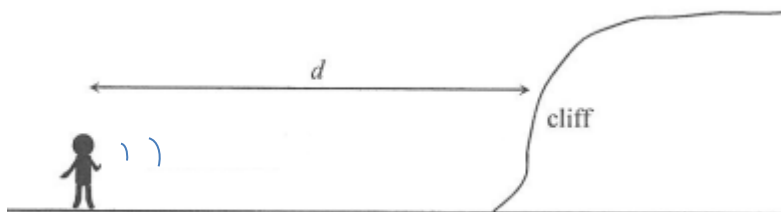
∴ (1) is incorrect.

The wavelength of an electromagnetic wave can be very long. For example, the wavelength of radio waves of frequency 1000 kHz is 300 m.

∴ (2) is not necessarily correct.

A

(i) Reflection of sound wave



Application of ultrasonic waves:

- Sonar for detecting the depth of sea bed
- Auto focusing in camera
- Ultrasonic cleaner
- Ultrasonic scanner
- To smash kidney and bladder stones
- To detect cracks in railway or machines

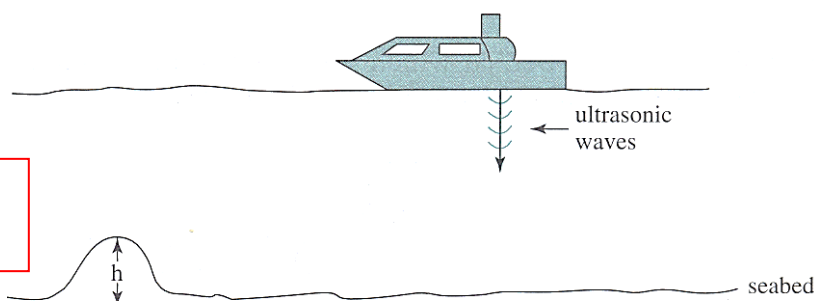
- Sound produces echo which is reflection of sound from a barrier.
- If the echo is heard after a **time interval** Δt , then the **distance** d between the observer and the reflector is related to the **speed of sound** v by

$$v = \frac{2d}{\Delta t}$$

- Ultrasounds are **sound waves** with frequency **greater than 20000 Hz**.
- **Ultrasound** is used for **sonar system** instead of audible sound since ultrasound has shorter wavelength so that it is less diffracted by obstacles and can be reflected by small objects.

14. The reflected waves are received 0.6 s after the emission of the ultrasonic waves (Velocity of sound in water is 1450 m s^{-1}).

(a) What is the depth of the ocean?



$$v = \frac{2d}{t}, d = \frac{vt}{2} = \frac{(1450)(0.6)}{2} = 435 \text{ m}$$

- (b) The ship then passes over a hump of the seabed. The time interval between the reflected waves and the transmitted waves is reduced to 0.56 s. Find the height of the hump.

$$v = \frac{2(d-h)}{t}, h = d - \frac{vt}{2} = 435 - \frac{(1450)(0.56)}{2} = 29 \text{ m}$$

15. A ship sends a pulse of ultrasound into the sea. 0.2 s later, the pulse reflected by a shoal of fish is received. Take the speed of sound in the sea as 1500 ms^{-1} . What is the distance between the ship and the shoal of fish?

- A. 150 m
B. 300 m
C. 600 m
D. 750 m

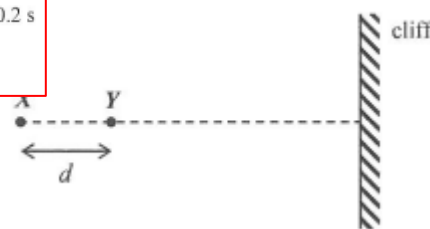
$$\text{A Distance} = \frac{1}{2}vt = \frac{1}{2} \times 1500 \times 0.2 = 150 \text{ m}$$

A

16. A boy claps his hands in front of a cliff at the position X and hears the echo 0.8 s later. He then walks a distance d towards the cliff and claps again at the position Y as shown in the above figure. This time the echo is heard 0.6 s later. Find d if the speed of sound in air is 330 m s^{-1} .

- A. 17 m
B. 33 m
C. 66 m
D. 132 m

When the boy walks a distance d , the echo is heard with an extra time of $(0.8 - 0.6) = 0.2 \text{ s}$
 $d = \frac{1}{2} \times 330 \times 0.2 = 33 \text{ m}$

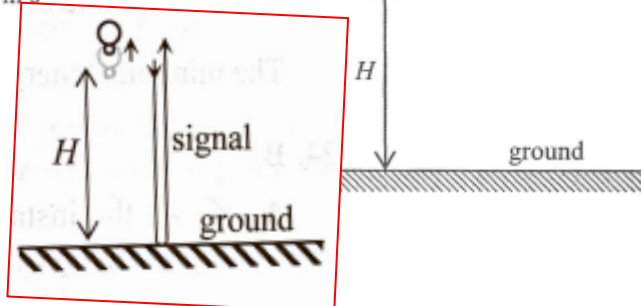


B

17. A balloon is rising at a uniform speed of 20 m s^{-1} . When the balloon is at an altitude H as shown, it sends a sound signal towards the ground. After 5 s, the balloon receives the echo of the signal. Estimate H . Given: speed of sound in air = 340 m s^{-1}

- A. 1600 m
B. 850 m
C. 800 m
D. 750 m

By $d = vt$
 $\therefore (H + H + 20 \times 5) = (340) \times (5)$
 $\therefore H = 800 \text{ m}$



C

18. 2014 DSE Q.17

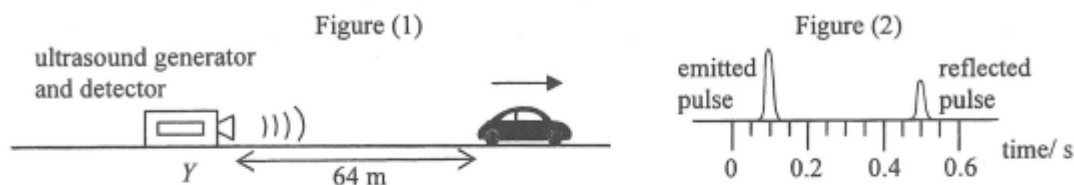


Figure (1) shows a car travelling with a uniform speed along a straight road away from a stationary ultrasound generator and detector at Y . When the car is 64 m from Y , the generator emits an ultrasound pulse towards the car. The pulse is then reflected back to the detector at Y and displayed on a CRO as shown in Figure (2). Estimate the speed of the car. Given: speed of ultrasound in air is 340 m s^{-1}

- A. 16 m s^{-1}
B. 20 m s^{-1}
C. 24 m s^{-1}
D. 32 m s^{-1}

Let v be the speed of the car.
The time taken by the pulse to reach the car = $(0.5 - 0.1) \div 2 = 0.2 \text{ s}$
 $\therefore 64 + v(0.2) = (340)(0.2)$
 $v = 20 \text{ m s}^{-1}$

B

19. Ultrasonic waves are used instead of audible sound waves to measure the depth of sea because
- (1) ultrasonic waves will not be interfered with by other sound waves in the water.
 - (2) ultrasonic waves can penetrate a greater depth of water with less spreading-out of waves.
 - (3) small-sized objects can also be located.

- A. (1), (2) and (3)
B. (1) and (2) only
C. (2) and (3) only
D. (1) only

✓ (1) Frequency of sound waves $\ll$ frequency of ultrasonic $\Rightarrow$ no interference of 2 waves
✓ (2) $f \uparrow \Rightarrow \lambda \downarrow \Rightarrow$ less diffraction $\Rightarrow$ less spreading out $\Rightarrow$ sound wave can penetrate a greater depth
✓ (3) $f \uparrow \Rightarrow \lambda \downarrow \Rightarrow$ much waves reflected $\Rightarrow$ can locate small objects

A

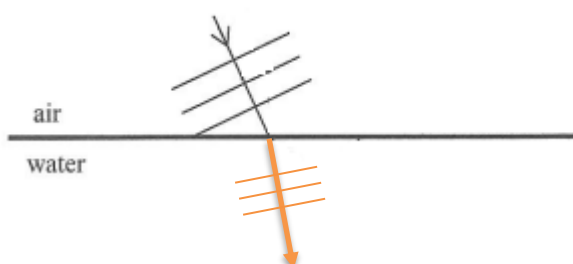
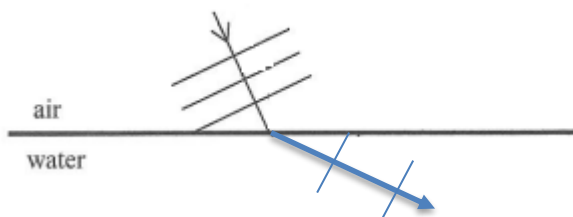
20. Submarines employ ultrasound instead of microwaves to detect obstacles in the sea. This is because

- A. wavelengths of ultrasound are shorter than those of microwaves.
B. ultrasound travels faster than microwaves in the sea.
C. microwaves are easily absorbed by sea water.
D. microwaves diffract too much in the sea.

D. C.
A × The wavelengths of ultrasound are longer than those of microwaves.
B × Ultrasound travels slower than microwaves in the sea.
C ✓ Since microwaves are easily absorbed by sea water, they cannot penetrate sea water.
D × (Continue from the explanation in A.)
Ultrasounds diffract more significantly than microwaves in the sea.

C

(ii) Refraction of sound



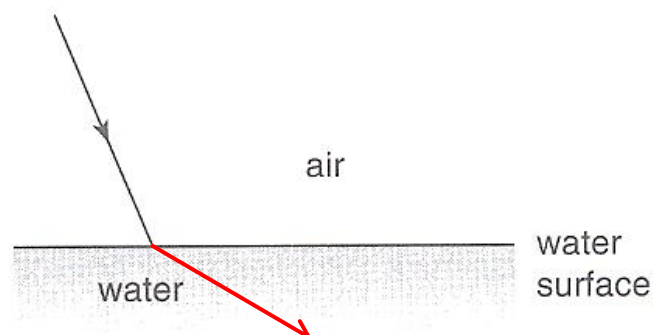
- Refraction of sound occurs as sound travels from one medium to another medium with **different speeds**.
- Frequency of the sound remains unchanged during the refraction.
- Relation between **speed, wavelength** and **angle** during refraction:
$$v \propto \lambda \propto \sin \theta$$
- **Total internal reflection** may occur if sound wave travels from **lower speed** (denser) medium to **higher speed** (less dense) medium and angle of incidence is larger than the critical angle.

21. A narrow beam of a sound wave travels into water. A swimmer under water hears the sound, but to a much lower loudness.

- Draw a diagram to show how the traveling direction of the sound wave changes. Explain briefly.
- State any changes in the wave properties of the sound.

- Denser $\rightarrow$ less dense
Bends away from normal

- Answer:
wavelength and wave speed increases
frequency no change



22. A train of ultrasound waves travels from air to the ground. The speeds of the ultrasound in air and the ground are 340 ms^{-1} and 5000 ms^{-1} respectively. The wavelength of the ultrasound in air is 1 cm. What is the wavelength of the ultrasound in the ground?

- 0.068 cm
- 3.4 cm
- 14.7 cm
- 500 cm

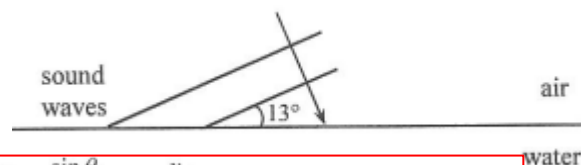
Since f remains unchanged, by $v = f\lambda$,

$$\begin{aligned} \frac{v_1}{\lambda_1} &= \frac{v_2}{\lambda_2} \\ \lambda_2 &= \frac{v_2}{v_1} \times \lambda_1 \\ &= \frac{5000}{340} \times 1 \\ &= 14.7 \text{ cm} \end{aligned}$$

C

23. Sound waves of frequency 1000 Hz travel from air to water as shown. If the incident plane wavefront makes an angle of 13° with the interface, find the angle of refraction and the wavelength of sound in water. (Given : speed of sound in air and that in water are 340 m s^{-1} and 1500 m s^{-1} respectively)

	angle of refraction	wavelength in water
A.	2.9°	7.7 cm
B.	2.9°	1.5 m
C.	83°	7.7 cm
D.	83°	1.5 m



D

$$\text{By } \frac{\sin \theta_{\text{air}}}{\sin \theta_{\text{water}}} = \frac{v_{\text{air}}}{v_{\text{water}}}$$

$$\therefore \frac{\sin 13^\circ}{\sin \theta_{\text{water}}} = \frac{340}{1500} \quad \therefore \theta_{\text{water}} = 83^\circ$$

$$\text{By } v_{\text{water}} = f \lambda_{\text{water}}$$

$$\therefore (1500) = (1000) \lambda_{\text{water}} \quad \therefore \lambda_{\text{water}} = 1.5 \text{ m}$$

24. Which of the following statements is/are correct?

- (1) The speed of sound in air depends on the temperature of the air.
- (2) When sound travels from air to water its frequency remains unchanged.
- (3) When sound travels from air to water its wavelength decreases.

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

Sound travels faster in hot air than in cold air.
 $\therefore$ (1) is correct.
 The frequency of a wave remains unchanged when it undergoes refraction.
 $\therefore$ (2) is correct.
 Sound travels faster in water than in air.
 By $v = f\lambda$, $v \uparrow$ and f unchanged $\Rightarrow \lambda \uparrow$
 $\therefore$ (3) is incorrect.

C

(iii) Diffraction of sound wave

- Sound has a very large degree of diffraction due to its long wavelength and hence we can hear sound round a corner.
- Sound wave has a much longer wavelength than light, thus the diffraction of sound is more significant than that of light.
- Diffraction of sound increase if
 - (1) wavelength of the sound wave increases
 - (2) size of the slit decreases

25. The diffraction of sound can be observed more easily in daily life than that of light because

- A. the energy transferred by sound is more.
- B. the wavelength of sound is larger.
- C. the speed of sound is lower.
- D. sound is a longitudinal wave.

B

B The degree of diffraction is large when the wavelength of the wave is comparable to the size of an obstacle. The wavelength of audible sound is between 0.017 m and 17 m while the wavelength of visible light is between 400 nm and 700 nm.
 $\therefore$ The diffraction of sound can be observed more easily in daily life.

Interference of sound waves



To observe interference of sound, 2 loudspeakers should be connected to the same signal generator to become 2 **coherent sources**. The frequency of the signal generator should be adjusted to give out sound of audible sound.

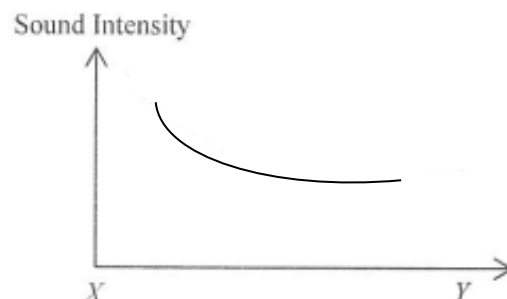
- When interference occurs:

(i) At points of constructive interference where $\Delta = n\lambda$, a loud sound is heard.

(ii) At points of destructive interference where $\Delta = (n + \frac{1}{2})\lambda$, a soft sound is heard.

As a result, alternative loud and soft sounds are heard across in front of the loudspeakers.

- The intensity along XY is always large and the loudness is always high. Since the path difference along XY is always zero, constructive interference occurs along XY. However, the sound intensity decreases gradually from X to Y as wave intensity decrease the distance.

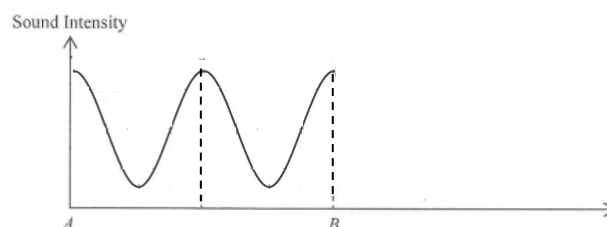


- Along AB, maximum and minimum intensity of sound can be observed alternately. The maximum is due to constructive interference. The minimum intensity is due to destructive interference.

- Reasons for non-zero intensity at minimum:

(i) There are background noises.

(ii) The destructive interference is not complete as 2 waves have different amplitudes due to different path lengths.



- If two sound sources emit sound of the same frequency but in anti-phase:

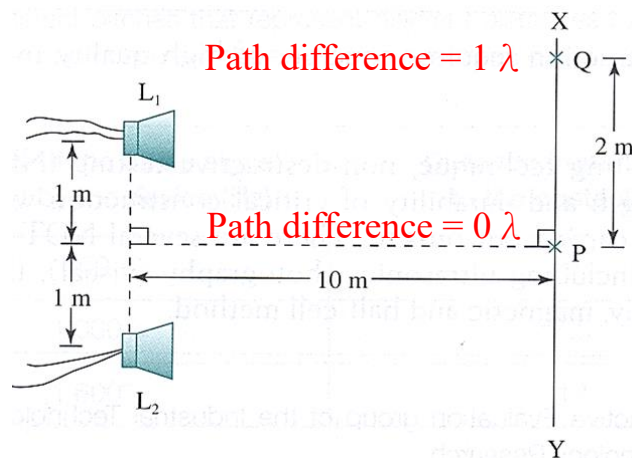
(i) the condition for the type of interference would reverse.

(ii) the central position would become a minimum.

26. (i) Describe the variation in the loudness heard by a man as he is walking along XY.

Alternate loud and soft sound

- (ii) Sound of maximum intensity is heard at P and Q. Find the frequency of the signal generator.
(Velocity of sound in air is 330 m s^{-1})



Answer : $L_1Q = \sqrt{10^2 + 1^2} = \sqrt{101} \text{ m}$ $L_2Q = \sqrt{10^2 + 3^2} = \sqrt{109} \text{ m}$
Path difference at Q = $\sqrt{109} - \sqrt{101} = 0.39 \text{ m} = \lambda$ ($\because$ the second constructive interference)

$$f = \frac{v}{\lambda} = \frac{330}{0.39} = 845 \text{ Hz}$$

27. As shown below, two loudspeakers emit coherent sound waves. A microphone receives sound of maximum loudness at P and sound of minimum loudness at Q. P is equidistant from the loudspeakers. Q is 3 m away from one loudspeaker and 2.76 m away from the other.

Which of the following can be the wavelength of the sound?

- (1) 0.12 m
(2) 0.16 m
(3) 0.48 m

- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

D Destructive interference occurs at Q.

$$\therefore \text{Path difference at Q} = \left(n + \frac{1}{2}\right)\lambda = 3 - 2.76 = 0.24 \text{ m}$$

$$\text{When } n = 0, \lambda = 0.48 \text{ m}$$

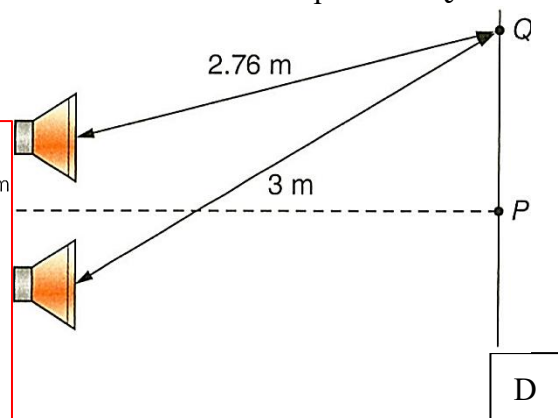
$$\text{When } n = 1, \lambda = 0.16 \text{ m}$$

$\therefore$ (2) and (3) are correct.

$$\text{If } \lambda = 0.12 \text{ m, path difference} = 2\lambda$$

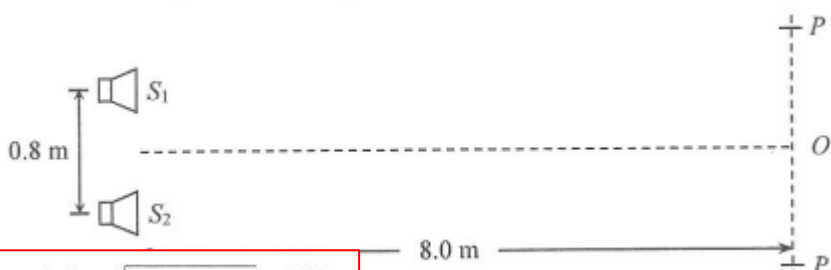
$\therefore$ Constructive interference occurs.

$\therefore$ (1) is incorrect.



28. S_1 and S_2 are two loudspeakers connected to a signal generator. The separation between S_1 and S_2 is 0.8 m. A student moves a microphone along a line PP' 8.0 m away from the loudspeakers and parallel to S_1S_2 . Loud sound is detected consecutively at P, O and P'. If PP' equals 2.0 m, estimate the wavelength of the sound produced by the loudspeakers.

- A. 5 cm
B. 10 cm
C. 15 cm
D. 20 cm



$$S_1P = \sqrt{(8)^2 + (1-0.4)^2} = 8.022 \text{ m}$$

$$S_2P = \sqrt{(8)^2 + (1+0.4)^2} = 8.122 \text{ m}$$

$$\text{Path difference at P} = 8.122 - 8.022 = 0.1 \text{ m}$$

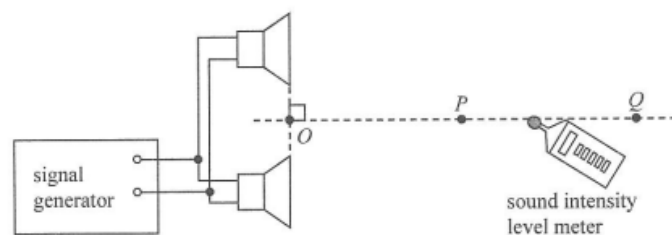
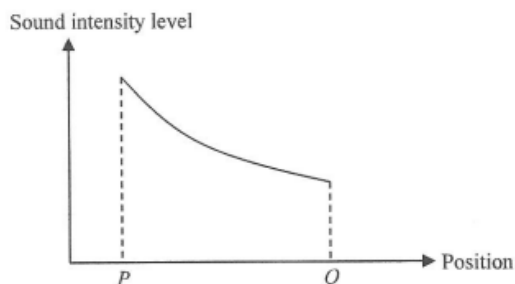
Since P is at the first maximum from O, path difference at P is 1λ .

$$\therefore \lambda = 0.1 \text{ m} = 10 \text{ cm}$$

違者必究

29. Two identical loudspeakers are connected in parallel to a signal generator, O is the midpoint between the loudspeakers. When a sound intensity level meter is moved from P to Q , the graph below shows the variation of the sound intensity level received with position.

A



Which of the following statements is/are correct ?

- (1) The sound intensity level at Q is smaller because the amplitude of the sound wave decreases as it is further away from the loudspeakers.
- (2) Constructive interference occurs at P while destructive interference occurs at Q .
- (3) The result of the experiment shows that sound is a wave.

- A. (1) only
B. (3) only
C. (1) & (2) only
D. (2) & (3) only

- ✓ (1) Sound intensity decreases with distance.
× (2) As the path difference at Q is zero, Q has constructive interference.
× (3) This experiment only shows that the sound intensity decreases with distance. To prove that sound is a wave, interference must be demonstrated. Alternate loud and soft sound should be shown to demonstrate the phenomenon of interference.

30. 2016 DSE MC Q.21

Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ . Which statement is **INCORRECT** ?

- A. Both PS_1 and PS_2 must be integral multiples of wavelength λ .
B. The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.

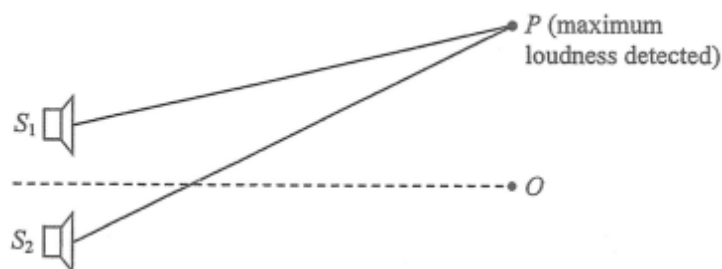
A

Note that maximum loudness indicates the occurrence of constructive interference and minimum loudness indicates the occurrence of destructive interference.

Since the sound waves from S_1 and S_2 are in phase and at point P maximum loudness is detected, the path difference $PS_2 - PS_1$ must be an integral multiple of wavelength λ .

- A. × Here is a counter example: Suppose $PS_1 = 4.7\lambda$ and $PS_2 = 5.7\lambda$. Then $PS_2 - PS_1$ is an integral multiple of λ but both PS_1 and PS_2 are not.
B. ✓ Since the path difference $PS_2 - PS_1$ may be $1\lambda, 2\lambda, \dots$, etc, its definite value cannot be determined from the information given.
C. ✓ Note that O is equidistant from the loudspeakers. O lies on an antinodal line and P lies on another. Since there is at least one nodal line between two antinodal lines, at least one point of minimum loudness can be detected between O and P .
D. ✓ If the sound waves from S_1 and S_2 are in antiphase, then minimum loudness will be detected when the path difference is an integral multiple of wavelength λ .

can be detected between O and P .
 P if the sound waves from S_1 and S_2 are in

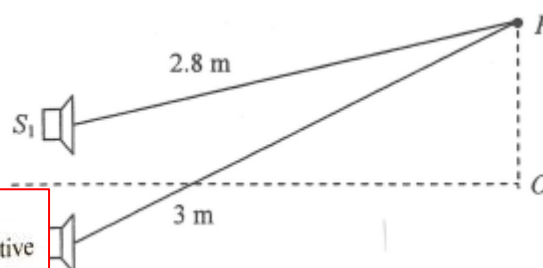


31. 2014 DSE MC Q.18

S_1 and S_2 are two loudspeakers connected to a signal generator but the sound waves produced by them are in anti-phase. Point O is equidistant from the loudspeakers while point P is at the distances shown in the figure from the loudspeakers. What type of interference occurs at O and P if the wavelength of the sound waves is 10 cm ?

C

- | | O | P |
|----|--------------|--------------|
| A. | destructive | constructive |
| B. | constructive | constructive |
| C. | destructive | destructive |



Let λ be the wavelength of the sound waves.
Since the sound waves produced by S_1 and S_2 are in anti-phase, destructive interference occurs when the path difference equals $0.5\lambda, 1.5\lambda, \dots$, etc and constructive interference occurs when the path difference equals $0\lambda, 1\lambda, \dots$, etc.

Since O is equidistant from the two loudspeakers, path difference at O is 0λ .

The path difference at $P = 3 - 2.8 = 0.2 \text{ m} = 2\lambda$

∴ Destructive interference occurs at both O and P .

違者必究

32. 2019 DSE MC Q.21

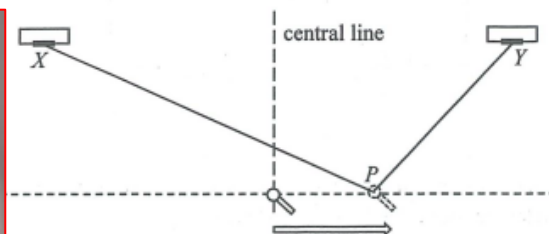
Two loudspeakers X and Y emit sound waves of frequency 500 Hz. A microphone is moved steadily along a line perpendicular to the central line as shown. It detects sound waves of maximum amplitude at the central line and the next maximum amplitude is detected at point P . Find $PX - PY$.

Given: speed of sound in air = 340 m s^{-1}

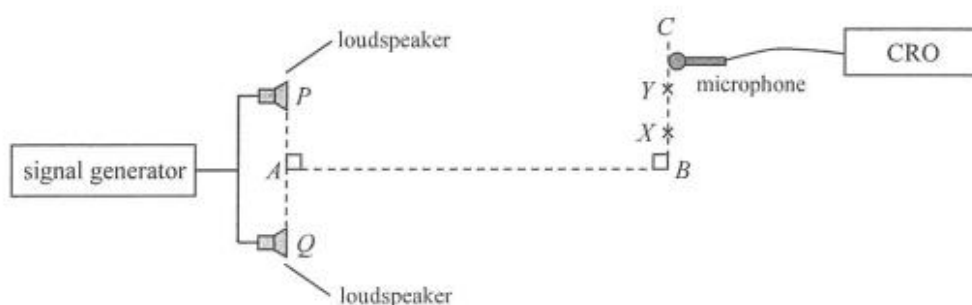
D

- A. 0.17 m
B. 0.34 m
C. 0.51 m
D. 0.68 m

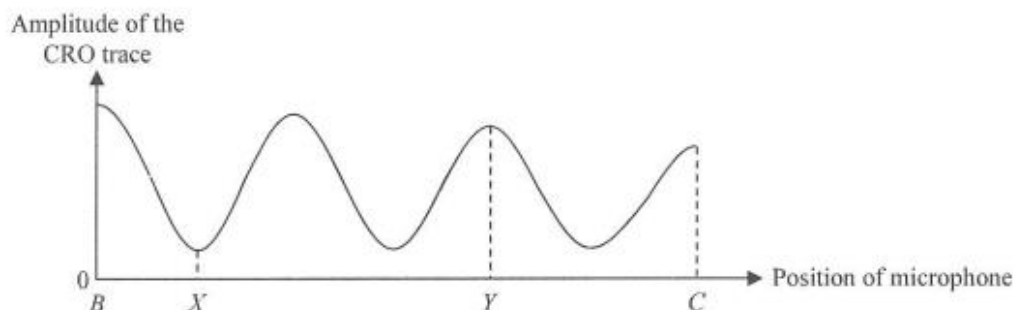
Let λ be the wavelength of the sound waves.
By $v = f\lambda$, $\lambda = 340 \div 500 = 0.68 \text{ m}$
Note that the path difference between the two loudspeakers at the central line is zero.
Since a maximum amplitude is detected at the central line and the next maximum amplitude is detected at point P , the path difference between the two loudspeakers at point P is λ .
 $\therefore PX - PY = 0.68 \text{ m}$



33.



In the above Figure, two identical loudspeakers P and Q are connected to a signal generator. Position A is the midpoint of PQ . A microphone connected to a CRO is moved along BC to measure the loudness of the sound. The amplitude of the CRO trace increases as the loudness of the sound detected increases. The Figure below shows the result.



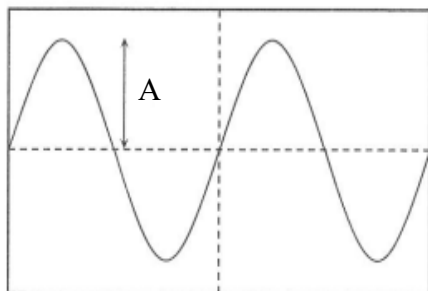
- (a) (i) Explain why the loudness of the sound varies at different positions along BC . (2 marks)
(ii) State ONE reason why the amplitude of the CRO trace is NOT zero at position X . (1 mark)
(b) If $PY = 5.10 \text{ m}$ and $QY = 5.78 \text{ m}$, find
(i) the path difference at position Y from P and Q ; (1 mark)
(ii) the wavelength of the sound. (2 marks)

(a) (i) Interference occurred. There are alternative constructive and destructive interferences occurs across BC . Hence, alternative maximum and minimum meter readings occur from B to C .	1 A 1 A
(ii) Background noise.	2 1 A 1
(b) (i) Path difference $= QY - PY$ $= 5.78 - 5.10$ $= 0.68 \text{ m}$	1 A 1
(ii) $2\lambda = 0.68$ $\lambda = 0.34 \text{ m}$	1 M 1 A 2

Musical notes

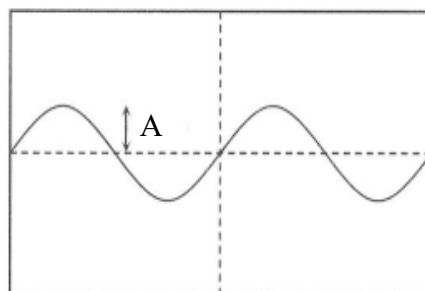
(i) Loudness

The loudness of a note depends on the amplitude of the sound wave. The **greater** the **amplitude**, the **larger** is the **loudness**.



$$A_1 > A_2$$

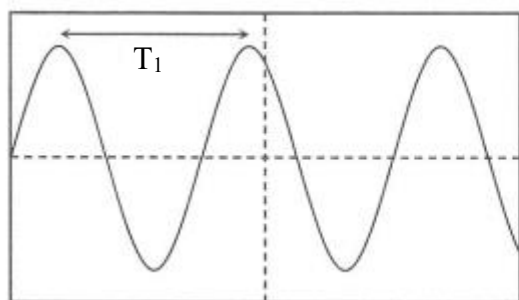
Louder



Softer

(ii) Pitch

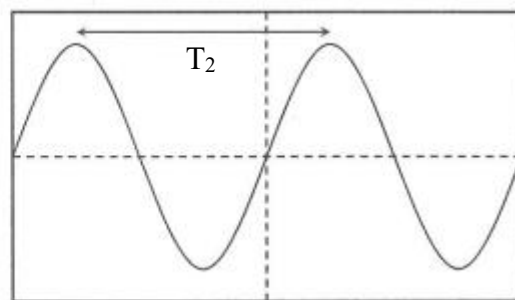
The pitch of a note depends on the frequency of the sound. The **higher** the **frequency**, the **higher** is the **pitch**.



Higher pitch

$$T_1 < T_2$$

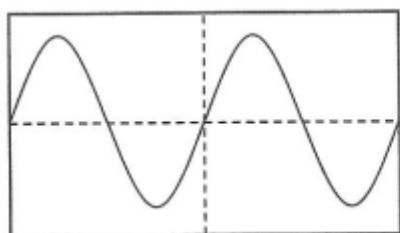
$$f_1 > f_2$$



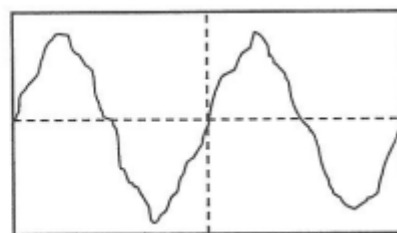
Lower pitch

(iii) Quality

Different musical instruments give sound of different quality. Quality of a sound depends on the waveform of the sound note. A tuning fork gives a pure sinusoidal waveform but other musical instruments give their characteristic irregular waveforms.



Tuning fork

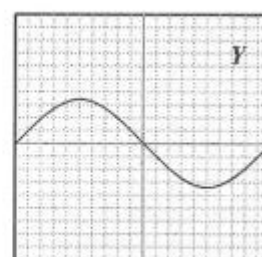
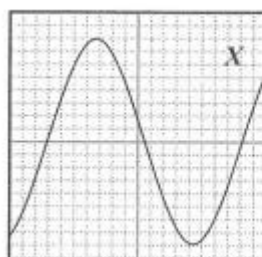


Other musical instrument

34. The diagrams above show the traces of two musical notes X and Y on an oscilloscope. Which of the following statements is/are true?

- (1) X has a higher pitch than Y .
- (2) The loudness of X is greater than that of Y .
- (3) X and Y are longitudinal waves.

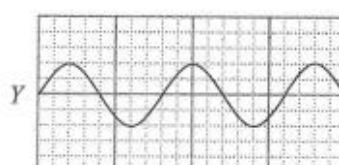
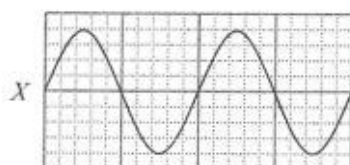
- A. (2) only
- B. (1) & (2) only
- C. (1) & (3) only
- D. (1), (2) & (3)



D

- ✓ (1) X has shorter period, thus it has higher frequency, therefore, X has higher pitch.
- ✓ (2) X has a greater amplitude, thus it has greater loudness
- ✓ (3) Both X and Y are musical notes, i.e. they are sound waves, thus they are longitudinal waves.

- 35.



A microphone and a CRO are used to detect the sound emitted by two tuning forks X and Y in turn. The figures above show the traces obtained, with the setting of the CRO unchanged. Find the ratio of the frequency of the sounds emitted by X to that of Y .

- A. 1 : 2
- B. 2 : 1
- C. 4 : 5
- D. 5 : 4

For a certain interval of time displayed in the CRO,

X completes 2 cycles but Y completes 2.5 cycles.

$$f_X : f_Y = 2 : 2.5 = 4 : 5$$

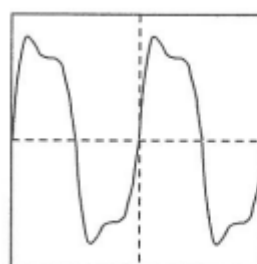
C

- 36.

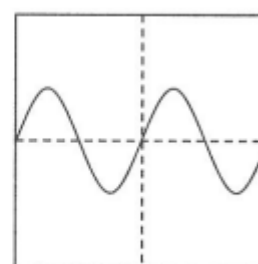
The above figures show the CRO displays of two musical notes X and Y . The settings of the CRO for the two notes are identical. Which of the following statements are correct?

- (1) Notes X and Y are of different qualities.
- (2) Note X is of a higher pitch than note Y .
- (3) Note X is louder than note Y .

- A. (1) & (2) only
- B. (1) & (3) only
- C. (2) & (3) only
- D. (1), (2) & (3)



Note X



Note Y

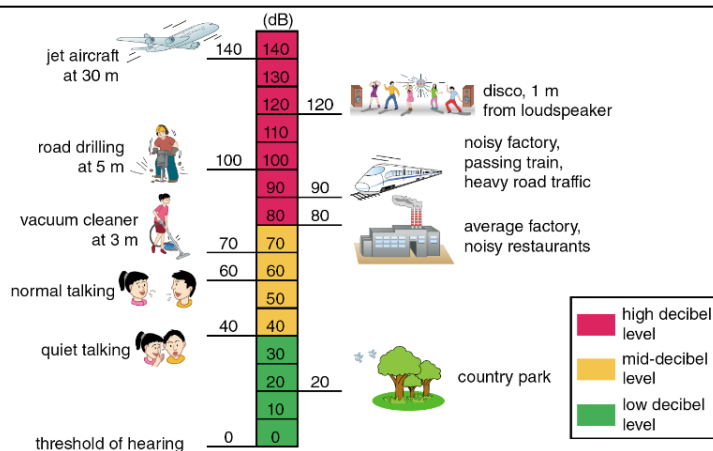
B

- ✓ (1) Notes X and Y are of different qualities as they have different waveforms.
- ✗ (2) Notes X and Y have the same pitch as they have the same frequency (same period).
- ✓ (3) Note X has larger amplitude than note Y , so note X is louder than note Y .

Sound intensity level

- Loudness of sound is indicated by the sound intensity level. It can be measured by sound intensity level meter.
- Decibel (dB)** is the unit of sound intensity level, h
- Threshold of hearing (0 dB) is the lowest sound intensity level that can be heard by human ear and does not mean 'no sound'.

Remarks: An increase in 10 dB means intensity of sound is 10 times the original



37. Which of the following statements is **incorrect**?

- A Noises above 140 dB causes permanent damage to the ear.
- B Noises at 130 dB can cause ear pain.
- C There is no vibration in the air at 0 dB.
- D Sound intensity level meters can be used to measure the loudness of noises.

C

38. What are the approximate noise intensity levels in the following environments?

- I. In a library
- II. In an office
- III. In a construction site

	I	II	III
A.	20 dB	40 dB	60 dB
B.	30 dB	60 dB	100 dB
C.	60 dB	90 dB	70 dB
D.	70 dB	60 dB	90 dB

B

39. Which of the following is **NOT** a typical sound intensity level that occurs in daily life ?

- A. 130 dB : when an airplane take-off
- B. 110 dB : at a rock concert
- C. 80 dB : having a normal conversation
- D. 30 dB : inside a library

- I. C
- A ✓ The sound intensity level of the sound produced by an airplane engine is typically between 130 dB and 140 dB.
 - B ✓ The sound intensity level of the sound at a rock concert is typically between 110 dB and 120 dB.
 - C ✗ The sound intensity level of a normal conversation is typically between 40 dB and 60 dB.
 - D ✓ The sound intensity level inside a quite indoor place is typically between 20 dB and 30 dB.

C